

Offline Python Coding Assistant

Qwen3-1.7B (w4a16) running on the Snapdragon X Elite NPU

Snapdragon AI Lab Build & Present Challenge

Built and tested remotely on Qualcomm Device Cloud (Snapdragon X Elite, SC8380XP, Windows 11)

Problem: students learn to code without reliable internet

- Cloud coding assistants need a connection, an account and often a subscription.
- Many students have patchy connectivity, or cannot send their code to a third-party server.
- A small model that runs fully on the laptop gives instant, private help for Python basics.
- **Use case: offline Python coding tutor (write, explain and fix short snippets).**

Why on-device, and why the NPU

Offline

No network after the one-time model download. Works in a classroom or on a train.

Private

Student code and prompts never leave the machine.

Efficient

NPU inference leaves CPU and GPU free and is built for low-power, sustained AI workloads.

What we built, and the honest two-model journey

Final stack (verified)

Model: Qwen3-1.7B w4a16, official Qualcomm bundle (1.6 GiB)

Runtime: GenieX CLI v0.7.0 (QAIRT 2.45)

Device: Qualcomm Device Cloud, CRD8380X (SC8380XP)

Load line: runtime=qairt compute=npu

How we got here

First attempt: Qwen2.5-Coder-1.5B exported to ONNX. It failed at NPU profiling, because the NPU path needs w4a16 quantization.

Switch: Qualcomm's prebuilt Qwen3-1.7B w4a16, which loads and runs on the NPU.

Measured on the NPU (real runs)

~40-42

tokens/s decode
(range 39-45)

~0.05 s

time to first token
(0.052-0.058 s)

364-733

tokens/s prompt
processing

Live demo (recorded on the device): reverse a string, binary search in 2 lines, Fibonacci generator.
All three answers were correct.

Indicative numbers from single short runs, not a formal benchmark.

Limitations and next steps

Limitations (be honest)

1.7B general model, not code-specialised.
A decorator prompt gave a WRONG answer in an earlier take.
Numbers come from a few short runs.
Tested on Qualcomm Device Cloud; I do not own a Snapdragon laptop.

Next steps

Try a code-tuned model once a w4a16 NPU bundle exists.
Run a proper benchmark set (correctness and latency).
Wrap GenieX in a simple local chat UI for students.
Package an offline installer for classrooms.